4.9 Holditch's Theorem

WORKSHEET

The question

STEP III 2011, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Check the theorem in the simplest case. If $\mathcal{C}$ is a circle of radius R, what curve does P trace, and what is the area of the ring?

STAGE 1 Area from coordinates

Ask yourself With $x = r\cos\theta$ and $y = r\sin\theta$, all functions of t, compute $x\dot{y} - y\dot{x}$, where dots denote $\frac{\mathrm{d}}{\mathrm{d}t}$.

STAGE 3 The area swept by OA

Ask yourself Apply $(*)$ to A. Expand $x_A\dot{y}_A - y_A\dot{x}_A$ and sort the terms by powers of a.

Pause and try Now find $[B]$. (*Hint:* B is obtained from the formula for A by replacing a with what?)

STAGE 4 Eliminating f

Ask yourself Both A and B go once round the same curve $\mathcal{C}$. What does that tell you about $[A]$ and $[B]$?

Try this next

1 **The area of an ellipse.** Use $(*)$ with $x = p\cos t$, $y = q\sin t$ to show that an ellipse with semi-axes p and q has area πpq.

2 **The circle case.** Prove the intersecting chords theorem: if two chords of a circle meet at P, the products of the lengths of their two segments are equal. Deduce Holditch's theorem for a circle.

3 **The midpoint.** Show that $f = \pi(a - b)$ means that when P is the midpoint of the rod, $f = 0$. What is the ring's area then, in terms of the rod's length L?

4 **A triangle.** HARDER Use $(*)$ to find the area of the triangle with vertices $(0,0)$, (x_1, y_1), (x_2, y_2), traversing its boundary, and show it equals $\frac{1}{2}|x_1y_2 - x_2y_1|$. This is the start of the "shoelace formula" for the area of any polygon.

Further reading

- **Mark Levi, *The Mathematical Mechanic*** (Princeton University Press). Uses physical reasoning, with rods, wheels and springs, to prove mathematical results, in the spirit of this section.
- **H. M. Schey, *Div, Grad, Curl, and All That*** (Norton). An informal introduction to vector calculus, including Green's theorem.
- **Hamnet Holditch, "Geometrical theorem"**, *Quarterly Journal of Pure and Applied Mathematics* 2 (1858). The original note, a single short paragraph with none of the fine print.

The full guided solution, and what the question leads to at university, is in Section 4.9:
https://beyondstep.keithmcnulty.org/ch04/holditch.html